FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
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FIITJEE

# JEE(Advanced)-2016

ALL INDIA INTEGRATED TEST SERIES

## ANSWERS, HINTS & SOLUTIONS

## PART TEST– IV
## (Paper - 1)

| Q. No. | PHYSICS | CHEMISTRY | MATHEMATICS |
|---|---|---|---|
| 1. | **D** | **C** | **C** |
| 2. | **B** | **B** | **A** |
| 3. | **D** | **A** | **C** |
| 4. | **B** | **A** | **D** |
| 5. | **B** | **A** | **A** |
| 6. | **C** | **D** | **A** |
| 7. | **C** | **C** | **D** |
| 8. | **C** | **D** | **B** |
| 9. | **C** | **D** | **A** |
| 10. | **A** | **A** | **C** |
| 11. | **A, C, D** | **A, B** | **A, B, C, D** |
| 12. | **B, C, D** | **A, B** | **A** |
| 13. | **B, C, D** | **A, C** | **B, D** |
| 14. | **A, B, C** | **A, B** | **A** |
| 15. | **A, C** | **A, B** | **A,C, D** |
| 1. | **2** | **2** | **1** |
| 2.. | **4** | **8** | **7** |
| 3. | **0** | **2** | **7** |
| 4. | **2** | **4** | **8** |
| 5. | **8** | **4** | **8** |

## HINTS AND SOLUTION

# Physics

## PART – I

### SECTION – A

1. $\frac{1}{e}=\frac{e^{-10\lambda t}}{e^{-\lambda t}}=\frac{1}{e^{-\lambda t+10\lambda t}}=\frac{1}{e^{9\lambda t}}$

$9\lambda t = 1$

$t = \frac{1}{9\lambda}$

2. Shift due to slab is $= t\left(1-\frac{1}{\mu}\right)=20\left(1-\frac{1}{2}\right)=10$ cm

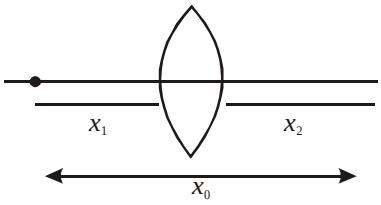
$\frac{1}{v}+\frac{1}{-(90)}=\frac{1}{-30}$

$\frac{1}{v}=\frac{1}{90}-\frac{1}{30}$

$\frac{1}{v}=\frac{1-3}{90}\Rightarrow v = -45$ m

so final image = 45 + 10(again shift) = 55 cm

3. Since object and image is real
then object distance $= -x_1$
Image distance $= x_2$




$x_0 = x_1 + x_2; \frac{1}{x_2}-\frac{1}{-x_1}=\frac{1}{f}$

$\Rightarrow\ x_1+x_2=\frac{x_1x_2}{f}\ \Rightarrow\ d=\frac{x_1(x_1-d)}{f}$

$\Rightarrow x_0=\frac{x_1^2}{x_1-f}\frac{dx_0}{dx_1}=0\ \Rightarrow x_1=2f$

which means $x_2 = 2f$
distance between object and image is 4f = 120 cm

5. $B.I_0 = K = I_1 + I_2 + \sqrt{2I_1I_2}\cos\phi\ \left(\phi=\frac{2\pi}{\lambda}\cdot\lambda\right)$

$k = 4I(I_1 = I_2 = I) \Rightarrow I = K/4$

when $\Delta x = \lambda/4$, $\phi = \frac{2\pi}{\lambda}\cdot\frac{\lambda}{4}=\frac{\pi}{2}$

$I' = I_1 + I_2 = \frac{K}{4}+\frac{K}{4}=\frac{K}{2}$

6. $f_{max}$(coherent) $= I + I + 2\sqrt{II}$

f(in coherent) = I + I = 2I

$\frac{f(\text{coherent})}{f(\text{incoherent})}=\frac{4I}{2I}=2$

7. $\sqrt{\nu}=a(z-b)$

$$\frac{k}{\sqrt{\lambda}} = a(z-b)$$

$$\frac{k}{\sqrt{\lambda_1}} = a(29-1) \qquad \text{....(i)}$$

$$\frac{k}{\sqrt{\lambda_2}} = a(15-1) \qquad \text{....(ii)}$$

from (i) and (ii) $\lambda_2 = 4\lambda_1 = 4\lambda$

11. $b \sin\theta = \left(n - \frac{1}{2}\right)\lambda$

$$\frac{b^2}{2D} = \left(n - \frac{1}{2}\right)\lambda$$

$\Rightarrow$ put n = 1, 2, 3, ……

$\Rightarrow \quad \lambda = \frac{b^2}{D}, \ \frac{b^2}{3D}, \ \frac{b^2}{5D}$

13. If power of the given lens is P

then $P_A = \frac{P}{2}; \ P_B = \frac{P}{2}; \ P_C = \frac{P}{2}$

15. Mosley's law :

$$\lambda \alpha \frac{1}{(z-1)} 2$$

$$\frac{\lambda_z}{\lambda_1} = \frac{(z_1-1)^2}{(z-1)^2}$$

$z_1 - 1 = (z-1)\,2$

$z_1 = 2z - 1$

$$\frac{\lambda_z}{\lambda_2} = \frac{1}{4} = \left(\frac{z_2 - 1}{z-1}\right)^2 .$$

## SECTION – C

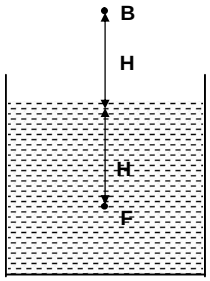
2. $$s_1 = \left[H + \frac{H}{\mu}\right]$$

$$s_2 = H + \mu H$$

$$\frac{s_1}{s_2} = \frac{1}{\mu}$$

$$\frac{s_1}{s_2} = \frac{1}{\mu} = 0.25$$

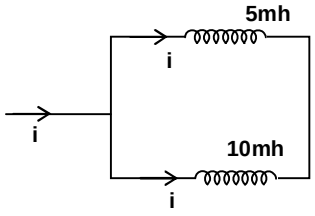
$$\mu = 4$$




5. $$i = \frac{10i}{15} = \frac{2i}{3} = \frac{x}{3}$$

$$i = \frac{20}{5} = 4$$

So, $x = 4 \times 2 = 8$



# Chemistry

## PART – II

### SECTION – A

2. $\frac{d_{bcc}}{d_{ccp}} = \frac{\text{packing efficiency of bcc}}{\text{packing efficiency of ccp}} = \frac{67.92}{74.02} = 0.918$

3. Except Ti(+IV) in $TiCl_4$ all other metal ions $Fe^{3+}$ in $FeCl_3$, $Cu^{2+}$ in $CuCl_2$ and $Mn^{2+}$ in $MnCl_2$ contains unpaired electrons and are coloured.

4. $Fe^{2+}$ = 0.07 mole in 0.1 mol of $Fe_{0.9}O$
So to balance charge, some of the place has been left vacant as $Fe^{2+}$ is replaced by $Sr^{4+}$.
$0.07 \times 2 = x \times 4$
$x = \frac{0.07 \times 2}{4} = 0.035$
Vacant place = $0.07 - 0.035 = 0.035\ N_A$

5. In diamond, all the carbon atoms are tetrahedrally bonded to one another so octahedral voids cannot be occupied. Also, the lattice points and the holes are occupied by carbon atoms. As a result the packing efficiency is less than 74%.

7. $4FeCl_3 + 3K_4[Fe(CN)_6] \longrightarrow \underset{\text{Prussian blue}}{Fe_4[Fe(CN)_6]_3} + 12KCl$

8. $SnCl_2 + HgCl_2 \longrightarrow SnCl_4 + \underset{(\text{white})}{Hg_2Cl_2}\downarrow$

$Hg_2Cl_2 + SnCl_2 \longrightarrow \underset{(\text{grey})}{Hg}\downarrow + SnCl_4$

9. $\frac{Y_A}{Y_B} = \frac{P_A / P_T}{P_B / P_T} = \frac{\chi_A . P_A^0}{\chi_B . P_B^0}$

$\Rightarrow \frac{Y_A}{\chi_A} = \frac{Y_B}{\chi_B}\left(\frac{P_A^0}{P_B^0}\right)$

Because $\frac{P_A^0}{P_B^0} > 1$

Hence $\frac{Y_A}{Y_B} > \frac{\chi_A}{\chi_B}$

10. $PBr_5(s) \longrightarrow [PBr_4]^+(s) + Br^-(s); \quad [i = 2]$

$2PCl_5(s) \longrightarrow [PCl_6]^- + [PCl_4]^+; \quad [i = 1]$

$\text{ratio} = \frac{2}{1} = 2$

11. $P_{total} = P_B^0 X_B + P_T^0 X_T$

$= 42 \times \frac{1}{3} + 36 \times \frac{2}{3} = 38$ mm.

$P_M Y_B = P_B^0 \times X_B$ ; Where $Y_B$ is the mole fraction of Benzene is vapour phase.

$38 \times Y_B = 14$

$Y_B = \frac{14}{38} = \frac{7}{19}$.

12. In the given reaction 'X' can be Au and $Ag_2S$.

$4Au + 8CN^- + 2H_2O + O_2 \longrightarrow 4Au(CN)_2^- + 4OH^-$

$2Ag_2S + 8CN^- + 2H_2O + O_2 \longrightarrow 4Ag(CN)_2^- + 4OH^- + 2S$

13. $Al^{+3} + 3e^- \rightarrow Al$

Number of moles of Al produced = $\frac{4.5}{27} = 0.166$

Number of moles of electrons used = $3 \times 0.166$

$H^+ + e^- \rightarrow \frac{1}{2}H_2$

Number of moles of $H_2$ produced = $\frac{1}{2} \times 3 \times 0.166$

Volume of $H_2$ produced = $1.5 \times 0.166 \times 22.4$ *l* = 5.57 = 5.6 *l*

15. Rochelle salt (sodium potassium tartrate) and bleaching powder (calcium chloride hypochlorite) are mixed salts. Calcium hypochlorite is a simple salt. Mohr's salt [$FeSO_4.(NH_4)_2SO_4.6H_2O$] is a double salt.

## SECTION – C

1. $0.7 = 2 \times K_f \times 0.2 \Rightarrow K_f = \frac{0.7}{0.4} = \frac{7}{4}$

$Hg(CN)_2 + mCN^- \longrightarrow Hg(CN)_{m+2}^{m-}$

| 0.1 | 0.2 | 0 |
|---|---|---|
| 0 | (0.2 – 0.1m) | 0.1 |

Now, Molarity of $K^+$ = 0.2 M

Molarity of $CN^-$ = (0.2 – 0.1m) M

Molarity of complex = 0.1 M

$0.53 = K_f (0.2 + 0.2 – 0.1m + 0.1) = K_f (0.5 – 0.1m)$

$\Rightarrow$ $0.53 = \frac{7}{4} (0.5 – 0.1m) \Rightarrow 2.1 = 3.5 – 0.7m$

$\Rightarrow$ $0.7m = 1.4 \Rightarrow m =$ **2**.

2. Number of moles of hypo = $\frac{MV}{1000}$

$= \frac{0.1\times 40}{1000} = 4\times 10^{-3}$

$\therefore$ Number of moles of $I_2 = 2\times 10^{-3}$

Mass of $I_2 = 2\times 10^{-3}\times 254$ g

$w = \frac{i\times t\times E}{96500}$

$2\times 10^{-3}\times 254 = \frac{i\times 9650\times 127}{96500}$

$i$ = 0.04 ampere

Current efficiency = $\frac{0.04}{0.50}\times 100$ = **8%**.

5. $Fe^{2+} + 2e^- \longrightarrow Fe(s); E^0 = 5V$ ...(1)

$Fe^{3+} + e^- \longrightarrow Fe^{2+}; \quad E^0 = 2V$ ...(2)

Equation (1) + (2):

$Fe^{3+} + 3e^- \longrightarrow Fe(s); E^0 = xV(let)$

Hence, $3x = 2\times 5 + 2$

$X = 4$.

# Mathematics

## PART – III

### SECTION – A

1. $\left(\frac{\vec{a}}{a^2}-\frac{\vec{b}}{b^2}\right)^2=\frac{a^2}{a^4}+\frac{b^2}{b^4}-\frac{2\vec{a}.\vec{b}}{a^2b^2}, \quad \{\because \vec{a}^2=a^2 \text{ etc.}\}$

$=\frac{a^2+b^2-2\vec{a}.\vec{b}}{a^2b^2}=\left(\frac{\vec{a}-\vec{b}}{ab}\right)^2.$

2. $(\vec{a}\times\vec{b})\times\vec{c}=\frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$

$\Rightarrow (\vec{a}.\vec{c})\vec{b}-(\vec{b}.\vec{c})\vec{a}=\frac{1}{3}|\vec{b}||\vec{c}|\vec{a}$

$\Rightarrow (\vec{a}.\vec{c})\vec{b}=\left\{(\vec{b}.\vec{c})+\frac{1}{3}|\vec{b}||\vec{c}|\right\}\vec{a}$

$\Rightarrow (\vec{a}.\vec{c})\vec{b}=|\vec{b}||\vec{c}|\left\{\cos\theta+\frac{1}{3}\right\}\vec{a}$

As $\vec{a}$ and $\vec{b}$ are not parallel, $\vec{a}.\vec{c}=0$ and $\cos\theta+\frac{1}{3}=0$

$\Rightarrow \cos\theta=-\frac{1}{3}$. Hence $\sin\theta=\frac{2\sqrt{2}}{3}$.

3. Let co-ordinates of foot of perpendicular are (a, b, c).

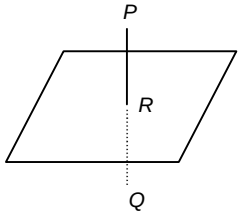
Since $\frac{a-11}{10}=\frac{b+2}{-4}=\frac{c+8}{-11}=k$

Then, $a=10k+11, b=-4k-2, c=-11k-8.$

Here all alternates give a =1, then k = –1. So foot is (1, 2, 3) and perpendicular distance between (1, 2, 3) and (2, – 1, 5) is $\sqrt{14}$.

Trick : The foot must satisfy the line, so obviously (1,2,3) satisfies. Now length of perpendicular is distance between the foot and the point i.e. $\sqrt{14}$. But students should remember here that it is not necessary that the line segment joining (2, – 1, 5) and (1, 2, 3) is perpendicular to the given line, so they should check. Therefore, direction ratios of the line joining (2, – 1, 5) and (1, 2, 3) are (– 1, 3, – 2) and that of given line are (10, – 4, – 11). Hence the lines are perpendicular.

5. Let Q be image of the point P(5,4,6) in the given plane, then PQ is normal to the plane. The direction ratios of PQ are 1,1,2. Since PQ passes through (5,4, 6) and has direction ratio 1,1,2;




Therefore, equation of PQ is

$\frac{x-5}{1}=\frac{y-4}{1}=\frac{z-6}{2}=r$, (say)

$\therefore\ x = r+5, y = r+4, z = 2r+6$

So, co-ordinates of Q be $(r+5, r+4, 2r+6)$

Let R be the mid point of PQ then co-ordinates of R are

$$\left(\frac{r+5+5}{2}, \frac{r+4+4}{2}, \frac{2r+6+6}{2}\right)$$

i.e., $\left(\frac{r+10}{2}, \frac{r+8}{2}, \frac{2r+12}{2}\right)$

Since R lies on the plane

$\therefore\ \frac{r+10}{2} + \frac{r+8}{2} + 2\left(\frac{2r+12}{2}\right) - 15 = 0$

$\Rightarrow\ r+10+r+8+4r+24-30 = 0$

$\Rightarrow\ 6r+12 = 0 \Rightarrow r = -2$

So, co-ordinates of Q are (3, 2, 2).

Trick : From option (a) , midpoint of (3, 2, 2) and (5, 4, 6) satisfies the equation of given plane.

7. Given $A = \begin{vmatrix} \sin(\theta+\alpha) & \cos(\theta+\alpha) & 1 \\ \sin(\theta+\beta) & \cos(\theta+\beta) & 1 \\ \sin(\theta+\gamma) & \cos(\theta+\gamma) & 1 \end{vmatrix}$

Operate $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$

$\therefore\ A = \{\cos(\theta+\gamma) - \cos(\theta+\alpha)\}$

$\{\sin(\theta+\beta) - \sin(\theta+\alpha)\} - \{\cos(\theta+\beta) - \cos(\theta+\alpha)\}\{\sin(\theta+\gamma) - \sin(\theta+\alpha)\}$

$= \sin(\beta-\gamma) - \sin(\beta-\alpha) - \sin(\alpha-\gamma)$,

which is independent of $\theta$.

8. $AB = \begin{bmatrix} \cos^2\theta & \sin\theta\cos\theta \\ \sin\theta\cos\theta & \sin^2\theta \end{bmatrix}\begin{bmatrix} \cos^2\varphi & \sin\varphi\cos\varphi \\ \sin\varphi\cos\varphi & \sin^2\varphi \end{bmatrix}$

$= \begin{bmatrix} \cos\theta\cos\varphi\cos(\theta-\varphi) & \cos\theta\sin\varphi\cos(\theta-\varphi) \\ \cos\theta\sin\varphi\cos(\theta-\varphi) & \sin\theta\sin\varphi\cos(\theta-\varphi) \end{bmatrix}$

$= \cos(\theta-\varphi)\begin{bmatrix} \cos\theta\cos\varphi & \cos\theta\sin\varphi \\ \cos\theta\sin\varphi & \sin\theta\sin\varphi \end{bmatrix}$

$= O$, $\left(\because \theta - \varphi = \frac{\pi}{2}\right)$.

9. Area $= \int_0^1 (1 + x - \sin x)dx + \int_1^2 (3 - x - \sin x)dx$

$= \frac{3}{2} + (\cos 1 - 1) + 3 - \frac{3}{2} + (\cos 2 - \cos 1) = 2 + \cos 2$

10. $x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$ or $\frac{dy}{dx} = \frac{y}{x} - \tan\left(\frac{y}{x}\right)$

It is homogeneous equation, hence put $y = vx$

we get, $v + x\frac{dv}{dx} = v - \tan v$

$\Rightarrow \int \cot v\, dv = -\int \frac{dx}{x}$

$\Rightarrow \log(x\sin v) = \log c$

$\Rightarrow x\sin\left(\frac{y}{x}\right) = c$.

13. $R_3 : R_3 - \alpha R_1 - R_2$

$$\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ 0 & 0 & -(a\alpha^2 + 2b\alpha + c) \end{vmatrix} = 0$$

$a\alpha^2 + 2b\alpha + c = 0$ or $b^2 = ac$

14. $P' = P^{-1}$ now $Q = PAP' = PAP^{-1}$

$Q^{2005} = \left(PAP^{-1}\right)^{2005} = PA^{2005}P^{-1}$

$\therefore \quad P'\left(Q^{2005}\right)P = P^{-1}\left(PA^{2005}P^{-1}\right)P$

$= A^{2005} = (I + B)^{2005}$

where $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ as $B^2 = 0$, we get $B^r = 0$, $\forall r \geq 2$ by Binomial theorem $A^{2005} = (I + B)^{2005}$

$= I + 2005\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$.

15. $$\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0$$

$(x+a)(x-b)(x+c) + (x+b)(x-a)(x-c) = 0$

$((x^2 + ac) + x(a+c))(x-b) + ((x^2 + ac) - (a+c)x)(x+b) = 0$

$2(x^2 + ac)x - 2bx(a+c) = 0$

$x = 0$ or $x^2 + ac - b(a+c) = 0$

## SECTION – C

1. $\vec{\alpha}.\vec{\beta} = \left(\frac{\hat{i}}{a} + \frac{4\hat{j}}{b} + b\hat{k}\right).\left(b\hat{i} + a\hat{j} + \frac{1}{b}\hat{k}\right) = \frac{b}{a} + \frac{4a}{b} + 1$

Hence $\frac{10}{5 + \vec{\alpha}.\vec{\beta}} = \frac{10}{5 + \frac{b}{a} + \frac{4a}{b} + 1} = \frac{10}{\frac{b}{a} + \frac{4a}{b} + 6}$

Then the minimum value of $\frac{b}{a} + \frac{4a}{b}$ is 4

$\therefore$ maximum value of $\frac{10}{5 + \vec{\alpha}.\vec{\beta}}$ is 1.

2. Equation of plane passing through the intersection of the planes $2x - 5y + z = 3$ and $x + y + 4z = 5$ is

$(2x - 5y + z - 3) + \lambda(x + y + 4z - 5) = 0$

$\Rightarrow (2 + \lambda)x + (-5 + \lambda)y + (1 + 4\lambda) - 3 - 5\lambda = 0$ ....(i)

which is parallel to the plane x + 3y + 6z = 1.

Then $\frac{2+\lambda}{1} = \frac{-5+\lambda}{3} = \frac{1+4\lambda}{6}$

Then, $\frac{2+\lambda}{1} = \frac{-5+\lambda}{3} = \frac{1+4\lambda}{6}$

$\therefore \lambda = \frac{-11}{2}$

from eq. (i),

$-\frac{7}{2}x - \frac{21}{2}y - 21z + \frac{49}{2} = 0$

$\therefore \; x + 3y + 6z = 7$

Hence, k = 7

3. $P(A \cap B) = P(A)\,P(B) \Rightarrow 1/6 = P(A)\,P(B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\Rightarrow 2/3 = P(A) + \frac{1}{6P(A)} - \frac{1}{6}$

$\Rightarrow 6(P(A))^2 - 5P(A) + 1 = 0 \Rightarrow P(A) = 1/2,\ P(B) = 1/3$

So, $8P(A) + 9P(B) = 4 + 3 = 7$

4. $f'(x) = \lim\limits_{h\to 0} \frac{f(x+h) - f(x+0)}{h} \Rightarrow \lim\limits_{h\to 0} \frac{f(x) + f(h) + h\sqrt{f(x)} - f(x) - f(0) - 0\sqrt{f(x)}}{h}$

$\Rightarrow \lim\limits_{h\to 0}\left(\frac{f(h) - f(0)}{h - 0}\right) + \sqrt{f(x)}$

$f'(x) = \sqrt{f(x)}$

$\int \frac{f'(x)}{\sqrt{f(x)}}dx = \int dx$

$2\sqrt{f(x)} = x + c$

$f(x) = \frac{x^2}{4}$

when $\alpha = 0$ area is minimum

required minimum area = $2\int_0^9 2\sqrt{y}\,dy \;\Rightarrow\; 4\left(\frac{y^{3/2}}{3/2}\right)_0^9$ = 72 sq. unit.

5. Put $y = vx$